

EDUCATION

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- University of Pittsburgh (Swanson School of Engineering/Frederick Honors Col.)** **Pittsburgh, PA**
 - B.S. - Computer Engineering (Autonomous Systems Focus); GPA: 3.99/4.00* *August 2022 - April 2026*
 - Courses:** Robotic Control, Data Structures and Algorithms, Embedded Processors, Microelectronics

PROFESSIONAL EXPERIENCE

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- ESTAT Actuation** **Pittsburgh, PA**
 - Robotics Developer Co-op* *May 2025 - November 2025*
 - Building and commissioning multiple robotic test stands expanding clutch testing capacity by 133%, covering mechanical assembly, electrical wiring, programming, and documentation for long-term use and potential sales to external customers.
 - Directing the assembly and delivery of 6 custom battery boxes for the Suzuki Moqba robot at the Japan Mobility Show. Overseeing quality control, leading debugging efforts to resolve heating issues, and ensuring a professional final build.
 - Testing new generation rotary clutch prototypes to improve reliability and evaluate longevity. Creating documentation explaining testing procedures, and using these prototypes to inform design decisions for next generation production clutches.
 - Carnegie Mellon University Robotics Institute** **Pittsburgh, PA**
 - Undergraduate Roboticist Researcher* *April 2023 - Present*
 - ZOË 2 Rover:** (tinyurl.com/zoe2rover)
 - Assisting in the development of software architecture and the robotic controller implementation for a novel 4-wheeled rover with a unique passive steering mechanism.
 - Enabling low-level software development for the rover using ROS2 and C++ to interface with motor controllers and encoders via CANOpen protocol.
 - Underwater Snake Robot:** (tinyurl.com/humrsCMU)
 - Implemented High-Frequency Injection methods in Brushless DC thrusters to achieve control at low/zero speeds thus reducing minimum speed by 80% allowing for reduced oscillations in station-keeping control loops.
 - Deployed station-keeping feature using AprilTags, IMU readings, and Nested PID Controllers to perform robot state-estimation underwater.
 - Conducted major repairs on the buoyancy module of the robot and assisted in continual electrical/hardware maintenance.
 - Apple's E-Waste Recycling Project:** (tinyurl.com/applecmu)
 - Labeled datasets consisting of 4000+ images for Machine Learning Models to detect screws in e-waste images.
 - Integrated ROS1 and Python packages to track AprilTags using a Realsense camera for localization of UR3 robotic arm.
 - Manufactured custom AprilTags using lasercutters and sheet metal manufacturing methods.

LEADERSHIP & ACTIVITIES

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- Robotics and Automation Society (University Rover Challenge)** **Pittsburgh, PA**
 - Chief Engineer/Integration Lead* *August 2022 - Present*
 - Leading a multidisciplinary team of 30 students in engineering a rover, coordinating integration efforts across mechanical, electrical, software, and bioscience teams. (tinyurl.com/roverimages)
 - Mentoring younger students in Onshape, KiCAD, and general manufacturing processes through hands-on building.
 - Securing and managing over \$7,000 in funding to support team development and future growth.
 - Proposed and spearheaded the development of a prototype robotic hand utilizing pneumatics. (tinyurl.com/hydraarm)

PUBLICATIONS

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- M. Nye, A. Raji, A. Saba, E. Erlich, R. Exley, **A. Goyal**, A. Matros, R. Misra, M. Sivaprakasam, M. Bertogna, D. Ramanan, and S. Scherer, **"BETTY Dataset: A Multi-modal Dataset for Full-Stack Autonomy,"** in Proc. IEEE Int. Conf. Robot. Autom. (ICRA), 2025. [Online]. Available: (arxiv.org/abs/2505.07266)

PROJECTS

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- Micromouse Robot:** Designed, assembled, and programmed a custom Micromouse robot through iterative PCB development, integrating motor control, IMU, IR sensors, and a display system with low-level embedded programming. Implemented PID control loops for navigation. Placed 3rd in competition at Stony Brook University IEEE conference. (tinyurl.com/goyalmm)
 - Bop It Project (KiCAD, Onshape, Microcontrollers):** Developed an interactive "Space Invaders Bop-It" reflex game using an ATmega328P IC, integrating analog inputs, audio-visual feedback, and custom embedded logic for randomized commands and progressively challenging gameplay. (tinyurl.com/spacebopit)
 - VoltVitals Embedded Board:** Designed and prototyped an ESP32 based circuit with web interface to measure home outlet voltage and report over Wi-Fi, collaborating with FirstEnergy engineers to enable early outage detection. (tinyurl.com/voltvitals)

SKILLS AND AWARDS

- **Software:** Linux, ROS1, ROS2, Github, Python, C++, C, MATLAB, ARM and RISC-V Assembly, CANOpen, Google Colab
- **Electrical:** Altium, KiCAD, LTSpice, Soldering, Arduino, ESP32
- **Mechanical:** Solidworks, Onshape, Milling, Lathe, MIG Welding, Laser Cutting, General Shop Tools
- **Awards:** Micromouse Competition 3rd place, SSOE Design Expo 1st place Art of Making, FSAE Innovation Award, Eagle Scout